

final_pipeline

February 7, 2026

```
[39]: import sys
import os

PROJECT_ROOT = os.path.abspath(os.path.join(os.getcwd(), ".."))
if PROJECT_ROOT not in sys.path:
    sys.path.insert(0, PROJECT_ROOT)

print("Project root added to PYTHONPATH:", PROJECT_ROOT)
```

Project root added to PYTHONPATH: c:\Users\acer\OneDrive\Desktop\iitbbsr

```
[ ]: import json
from tasks.task1_explain import task1_run, load_conversations
from context.context_manager import ContextManager
from tasks.task2_multiturn import answer_followup
from evaluation.id_recall import id_recall_from_explanation
from evaluation.faithfulness import faithfulness
```

```
[40]: from tasks.task1_explain import task1_run, load_conversations

DATA_PATH = "../data/processed/normalized_conversations.json"

conversations = load_conversations(DATA_PATH)

print(f"Loaded {len(conversations)} normalized conversations")
```

Loaded 5037 normalized conversations

```
[41]: import json
query = "Why did escalation occur?"

outcome, explanation = task1_run(
    query=query,
    data_path=DATA_PATH,
    use_embeddings=False # deterministic mode
)

print("=== TASK 1 OUTPUT ===")
print("Outcome:", outcome)
```

```
print(json.dumps(explanation.model_dump(), indent=2))
```

=== TASK 1 OUTPUT ===

Outcome: escalation - repeated service failures

```
{
  "outcome": "escalation - repeated service failures",
  "call_ids": [
    "7034-5430-2980-5483",
    "2801-6816-9124-1355",
    "3883-8555-9747-5746",
    "4967-8141-1813-8658",
    "3631-7868-1079-2880",
    "5405-1959-8150-2620",
    "6442-9317-2833-5263",
    "4471-5440-7427-7835",
    "6338-1986-7723-6070",
    "2717-1088-7257-7370",
    "9319-3940-9382-2387",
    "4999-7266-1895-5583",
    "3925-2095-7371-7901",
    "8584-4627-1759-3846",
    "8625-9811-3321-2163",
    "3925-3887-6255-8171",
    "3263-1585-8501-2701",
    "7111-8235-8458-4031",
    "9361-3226-5280-7468",
    "3578-4645-2490-5752",
    "9036-1648-4300-2522",
    "5186-2494-1700-6696",
    "7888-5665-2083-7600",
    "7552-6299-5982-6654",
    "1388-4256-8564-5555",
    "7946-2197-7898-4045",
    "9295-9855-5231-5046",
    "3528-5080-2778-7765",
    "6464-2429-2502-2616",
    "6625-3264-7615-2421",
    "2990-4204-4258-9488",
    "3853-4509-5928-6824",
    "9065-7143-5542-3940",
    "3425-7410-3440-7404",
    "3605-5214-9667-1936",
    "8144-4002-9143-6860",
    "7522-7011-4390-3472",
    "8167-2369-5496-4527",
    "5769-2119-7686-5558",
    "3319-1846-4085-2506",
    "9924-8351-1163-6855",
```

"6193-6676-9766-8161",
"5780-6506-8284-7747",
"6389-2133-4705-3957",
"6906-5754-3051-6961",
"2508-2921-7136-1365",
"6091-7333-4922-5653",
"1519-8313-1175-2744",
"1540-3642-8103-4857",
"9010-9788-2138-9621",
"2962-7737-7315-3742",
"9380-8564-2386-5850",
"2917-9389-6995-8046",
"4814-5375-3288-1200",
"7668-6398-3755-9343",
"6957-5242-1212-8562",
"5449-4261-3967-3740",
"6614-5759-9339-5398",
"6260-7488-2858-8247",
"8216-7621-8074-8507",
"1294-4560-6142-7703",
"8716-9296-8963-7931",
"3234-7094-1876-7539",
"9237-9924-3521-5157",
"4078-8032-5849-5035",
"6084-4943-4686-7826",
"9035-2173-8628-5445",
"9422-4550-4830-7307",
"8952-6892-2685-1452",
"4580-2907-7283-1968",
"7374-8812-5389-7412",
"1081-8483-5301-8677",
"7824-5101-2953-1240",
"7754-2394-5205-7020",
"9024-4171-3552-9634",
"4245-4349-4756-9570",
"4069-5699-2350-2526",
"5822-3107-8903-6566",
"5841-9475-2828-5120",
"8915-4457-6507-4260",
"7461-4984-1291-5153",
"5353-1811-9325-3323",
"4887-7630-1942-7697",
"3466-9954-4417-1756",
"1261-3470-8048-1661",
"2552-3995-5304-1163",
"9023-9433-9878-6116",
"5244-3362-1116-3682"

],

```

"factors": [
  {
    "name": "customer_frustration",
    "description": "Customer expresses frustration or dissatisfaction, often
indicating service failure or unresolved issues.",
    "evidence": [
      {
        "call_id": "7034-5430-2980-5483",
        "turn_id": 2,
        "speaker": "customer",
        "text": "I've been trying to resolve an issue for three weeks now and
I'm not getting any real assistance. My mobile app login keep failing.",
        "factor": "customer_frustration"
      },
      {
        "call_id": "7034-5430-2980-5483",
        "turn_id": 5,
        "speaker": "agent",
        "text": "I understand your frustration. Can you provide your account
number so I can review the history?",
        "factor": "customer_frustration"
      },
      {
        "call_id": "7034-5430-2980-5483",
        "turn_id": 8,
        "speaker": "customer",
        "text": "Fine, but make it quick. I've wasted enough time on this.",
        "factor": "customer_frustration"
      },
      {
        "call_id": "7034-5430-2980-5483",
        "turn_id": 16,
        "speaker": "customer",
        "text": "I still want to speak with a supervisor. This has been
incredibly frustrating and I want to file a formal complaint.",
        "factor": "customer_frustration"
      },
      {
        "call_id": "2801-6816-9124-1355",
        "turn_id": 4,
        "speaker": "customer",
        "text": "The account number is 7729-4483-2910. But I shouldn't have to
explain this again. This is the third person I've talked to.",
        "factor": "customer_frustration"
      },
      {
        "call_id": "2801-6816-9124-1355",
        "turn_id": 5,

```

```

        "speaker": "agent",
        "text": "I completely understand. Before I transfer you, let me
quickly review your account to brief them on the situation. That way you won't
have to explain everything again.",
        "factor": "explicit_escalation_request"
    },
    {
        "call_id": "5244-3362-1116-3682",
        "turn_id": 16,
        "speaker": "customer",
        "text": "I still want to speak with a supervisor. This has been
incredibly frustrating and I want to file a formal complaint.",
        "factor": "explicit_escalation_request"
    },
    {
        "call_id": "5244-3362-1116-3682",
        "turn_id": 17,
        "speaker": "agent",
        "text": "Absolutely, you have every right to speak with a supervisor.
Let me get your two-factor authentication verified first so we can prevent any
further issues, and then I'll transfer you directly to my supervisor.",
        "factor": "explicit_escalation_request"
    }
],
"strength": 1.0
}
],
"causal_chain": [
    "Repeated failures without resolution increased frustration.",
    "Lack of notification/communication increased perceived negligence.",
    "High-stakes urgency intensified the need for immediate action.",
    "Customer explicitly requested escalation to a supervisor/manager."
],
"summary": "For outcome 'escalation - repeated service failures', the
conversations show a progression where service issues and interaction dynamics
(e.g., frustration, repetition, lack of communication, urgency) lead to
escalation behaviors. All claims are grounded in quoted turn-level evidence
spans."
}

```

```

[42]: from context.context_manager import ContextManager
context_manager = ContextManager()
state = context_manager.initialize(explanation)

print("=== CONTEXT INITIALIZED ===")
print("Active Outcome:", state.active_outcome)
print("Active Call IDs:", state.active_call_ids)

```

```
print("Pinned Evidence Count:", len(state.pinned_evidence))
```

=== CONTEXT INITIALIZED ===

Active Outcome: escalation - repeated service failures

Active Call IDs: ['7034-5430-2980-5483', '2801-6816-9124-1355',
'3883-8555-9747-5746', '4967-8141-1813-8658', '3631-7868-1079-2880',
'5405-1959-8150-2620', '6442-9317-2833-5263', '4471-5440-7427-7835',
'6338-1986-7723-6070', '2717-1088-7257-7370', '9319-3940-9382-2387',
'4999-7266-1895-5583', '3925-2095-7371-7901', '8584-4627-1759-3846',
'8625-9811-3321-2163', '3925-3887-6255-8171', '3263-1585-8501-2701',
'7111-8235-8458-4031', '9361-3226-5280-7468', '3578-4645-2490-5752',
'9036-1648-4300-2522', '5186-2494-1700-6696', '7888-5665-2083-7600',
'7552-6299-5982-6654', '1388-4256-8564-5555', '7946-2197-7898-4045',
'9295-9855-5231-5046', '3528-5080-2778-7765', '6464-2429-2502-2616',
'6625-3264-7615-2421', '2990-4204-4258-9488', '3853-4509-5928-6824',
'9065-7143-5542-3940', '3425-7410-3440-7404', '3605-5214-9667-1936',
'8144-4002-9143-6860', '7522-7011-4390-3472', '8167-2369-5496-4527',
'5769-2119-7686-5558', '3319-1846-4085-2506', '9924-8351-1163-6855',
'5446-9855-3745-5784', '2278-2074-7495-8931', '2507-4864-2826-9884',
'7939-8853-2604-7320', '1924-8055-3750-6871', '1268-1020-3396-4901',
'9767-1897-3358-7252', '4333-7573-2943-9767', '1279-2604-4038-2594',
'3970-3828-1547-6943', '6333-4287-5565-2256', '2430-2298-2406-2095',
'9702-6826-2103-3733', '5562-6174-2571-9758', '4170-9507-9581-3369',
'3636-3502-3579-3641', '5403-7602-6638-8625', '9739-8892-6173-8705',
'1147-2303-5639-3789', '4407-3406-4003-6396', '6037-1189-7358-2962',
'9479-5415-5448-1945', '2599-4375-7221-5028', '1737-4975-4714-1298',
'2499-5325-9436-1435', '1191-3030-7742-5712', '5348-2070-1304-7386',
'6170-9665-5558-9818', '3283-5939-1125-4945', '9094-6131-5416-2004',
'6925-7843-6811-8098', '6181-4413-7724-7069', '6971-6872-7559-5094',
'9733-6344-8137-4785', '3714-9375-4956-5012', '6330-1131-2887-6227',
'8566-7297-5629-7341', '2217-8080-1649-6972', '3573-8790-9686-2581',
'1476-2339-5153-5295', '4790-2901-4157-3635', '1266-8269-6201-3683',
'1148-6899-1899-8115', '4998-9224-6842-8524', '9298-7389-8917-7383',
'3864-9288-4669-4638', '3258-2284-3769-6046', '6676-9385-5929-3212',
'7771-3832-5326-3365', '8420-9878-7447-5372', '4296-2682-1344-3926',
'6550-5775-8931-5849', '6029-2509-2015-1410', '2307-4311-3663-9783',
'3668-3420-3007-2389', '1783-9585-5540-5657', '4236-8890-8212-6303',
'5790-4148-7611-7524', '9210-9501-7892-4841', '1042-8261-1925-4264',
'2964-5551-3233-2464', '9719-7077-2927-9652', '6801-5289-1968-1047',
'8887-5278-8455-6513', '5041-8394-3878-2286', '3748-9351-1393-3728',
'3161-7824-9805-7606', '4282-4271-6056-9823', '6007-1604-1973-4430',
'3828-6784-1391-8824', '2679-6397-1633-2644', '4717-5765-6910-7909',
'8475-1230-5838-4957', '9107-1229-6371-5165', '3540-9030-1703-7485',
'9808-1463-6569-8600', '4271-7461-6691-1250', '6664-9672-2872-6994',
'6747-9112-7781-3874', '3474-5220-1228-3459', '3675-8067-7227-5984',
'5482-6580-4276-8878', '3719-5615-7150-4708', '7813-5250-5173-5897',
'2664-5177-4220-7696', '7230-3385-4522-8358', '2214-5591-1761-4959',
'3656-4360-7913-3583', '6613-2013-8329-9685', '3893-2990-4322-8394',

```
'6570-2571-8509-4147', '9856-1246-4474-7529', '5153-1985-4030-8257',
'1181-2133-8621-3866', '7439-5959-3289-5589', '9242-8320-4699-4071',
'1019-5034-8625-5277', '5798-1198-4201-1549', '8873-1142-7348-3790',
'4510-6622-2241-7098', '9728-4326-7364-3155', '2658-1641-8835-1027',
'2384-7582-4852-1648', '9280-4447-5714-4103', '6778-2900-2362-8605',
'6108-6515-5640-3092', '5802-9540-5222-4503', '7747-9927-9246-2846',
'5812-6792-4085-5897', '4645-8646-6624-8207', '4255-4937-6587-8528',
'4677-1723-1537-9067', '7624-2134-6787-6673', '7938-3799-4511-4255',
'2735-1406-6816-3770', '3919-3145-5703-7868', '9391-7266-1346-7002',
'5077-9782-9796-3266', '9858-7660-8845-5461', '9806-6502-1700-4953',
'4320-6673-8051-5876', '8651-9624-5021-9802', '2930-7897-6555-8267',
'1384-2625-2556-7586', '9241-7020-1248-6658', '7442-9234-3044-8926',
'1859-6631-3494-7017', '4563-8350-5001-8653', '4800-7018-4624-1857',
'1780-4069-7653-5083', '5375-6427-1663-8046', '7676-7790-1980-2868',
'1730-1729-4086-4502', '3701-6803-9085-9784', '2506-1813-2516-9541',
'7477-4569-7509-1808', '7096-9078-4281-5083', '4421-4793-2010-8351',
'8223-7262-3751-4834', '4056-8679-7531-3535', '8091-5377-8818-1386',
'6203-7894-9839-1849', '6694-5414-6092-7828', '1592-2984-7820-6849',
'6193-6676-9766-8161', '5780-6506-8284-7747', '6389-2133-4705-3957',
'6906-5754-3051-6961', '2508-2921-7136-1365', '6091-7333-4922-5653',
'1519-8313-1175-2744', '1540-3642-8103-4857', '9010-9788-2138-9621',
'2962-7737-7315-3742', '9380-8564-2386-5850', '2917-9389-6995-8046',
'4814-5375-3288-1200', '7668-6398-3755-9343', '6957-5242-1212-8562',
'5449-4261-3967-3740', '6614-5759-9339-5398', '6260-7488-2858-8247',
'8216-7621-8074-8507', '1294-4560-6142-7703', '8716-9296-8963-7931',
'3234-7094-1876-7539', '9237-9924-3521-5157', '4078-8032-5849-5035',
'6084-4943-4686-7826', '9035-2173-8628-5445', '9422-4550-4830-7307',
'8952-6892-2685-1452', '4580-2907-7283-1968', '7374-8812-5389-7412',
'1081-8483-5301-8677', '7824-5101-2953-1240', '7754-2394-5205-7020',
'9024-4171-3552-9634', '4245-4349-4756-9570', '4069-5699-2350-2526',
'5822-3107-8903-6566', '5841-9475-2828-5120', '8915-4457-6507-4260',
'7461-4984-1291-5153', '5353-1811-9325-3323', '4887-7630-1942-7697',
'3466-9954-4417-1756', '1261-3470-8048-1661', '2552-3995-5304-1163',
'9023-9433-9878-6116', '5244-3362-1116-3682']
```

Pinned Evidence Count: 11364

```
[43]: from tasks.task2_multiturn import answer_followup
followup_queries = [
    "Which factors contributed?",
    "Which turns caused escalation?",
    "Why did this happen?"
]

print("=== TASK 2 FOLLOW-UPS ===")

for q in followup_queries:
    context_manager.log_query(q, "followup")
```

```
response = answer_followup(state, q)
print(f"\nFOLLOW-UP QUESTION: {q}")
print(json.dumps(response, indent=2))
```

=== TASK 2 FOLLOW-UPS ===

FOLLOW-UP QUESTION: Which factors contributed?

```
{
  "category": "factors",
  "active_outcome": "escalation - repeated service failures",
  "factors": [
    {
      "name": "customer_frustration",
      "description": "Customer expresses frustration or dissatisfaction, often
indicating service failure or unresolved issues.",
      "evidence_count": 2688
    },
    {
      "name": "repetition_without_resolution",
      "description": "Customer indicates they repeated the issue multiple times
without resolution, increasing escalation likelihood.",
      "evidence_count": 1443
    },
    {
      "name": "lack_of_proactive_communication",
      "description": "Customer indicates they were not notified / informed,
which can trigger blame and escalation.",
      "evidence_count": 490
    },
    {
      "name": "high_stakes_urgency",
      "description": "Time-sensitive or high-impact consequences raise emotional
intensity and escalation risk.",
      "evidence_count": 2436
    },
    {
      "name": "explicit_escalation_request",
      "description": "Customer explicitly requests a
supervisor/manager/executive transfer.",
      "evidence_count": 4307
    }
  ]
}
```

FOLLOW-UP QUESTION: Which turns caused escalation?

```
{
  "category": "evidence",
  "active_outcome": "escalation - repeated service failures",
```

```
}
```

```
[44]: from evaluation.id_recall import id_recall_from_explanation
gold_call_ids = [
    c.call_id for c in conversations if c.outcome == outcome
]

idrecall = id_recall_from_explanation(explanation, gold_call_ids)

print("=== EVALUATION ===")
print("IDRecall:", idrecall)
```

```
=== EVALUATION ===
IDRecall: 1.0
```

```
[45]: from evaluation.faithfulness import faithfulness
faith = faithfulness(explanation, conversations)

print("Faithfulness:", faith)
```

```
Faithfulness: 1.0
```

```
[46]: print("Notebook execution completed successfully.")
```

```
Notebook execution completed successfully.
```

```
[48]: from evaluation.relevancy import relevancy

rel = relevancy(state, user_query, system_response)

print("Relevancy:", rel)
```

```
Relevancy: 1.0
```

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